

Chapter 1 Differentiable Manifolds

1.1 Differentiable Manifolds

1.1.1 Pseudo Group Axioms

Definition 1.1.1.1 A pseudogroup of transformations on a topological space S is a set Γ of transformations satisfying the following axioms:

- Each $f \in \Gamma$ is a homeomorphism of an open set of S onto another open set of S .

$$f_{\cong} : U \rightarrow V; U, V \in \text{Open}(S)$$

- If $f \in \Gamma$, then the restriction of f to an arbitrary open subset of domain of f is in Γ . (And it is still a homeomorphism)

$$f \in \Gamma \Rightarrow f|U \in \Gamma$$

- Let $U = \cup U_i$. Then,

$$f|U_i \in \Gamma \Rightarrow f \in \Gamma$$

- For every open U ,

$$id_U \in \Gamma$$

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$$f \in \Gamma \Rightarrow f^{-1} \in \Gamma$$

- $f \in \Gamma$ is a homeomorphism of U onto V , $f' \in \Gamma$ is a homeomorphism of U' onto V' , then

$$\Gamma \ni f' \circ f : f^{-1}(V \cap U') \rightarrow f'(V \cap U')$$

♣ **Example 1.1.1.2** (Pseudogroup $\Gamma^r(\mathbb{R}^n)$) Define:

$$\Gamma^r(\mathbb{R}^n) := \{f \in C^r(U, V) | U, V \in \text{Open}(\mathbb{R}^N)\}$$

There is an inclusion

$$\Gamma^{r+1}(\mathbb{R}^n) \hookrightarrow \Gamma^r(\mathbb{R}^n)$$

♣ **Example 1.1.1.3** Furthermore, Pseudogroup

$$\mathcal{O}(\Omega)$$

1.1.2 Atlas

Definition 1.1.2.1 An **atlas** of a topological space M is a family of **charts** (U_i, φ_i) which satisfies:

- Each U_i is an open set of M , and

$$\cup U_i = M$$

- $\varphi_i : U_i \rightarrow S$ is a homeomorphism onto its image;
- If $U_i \cap U_j \neq \emptyset$, then

$$\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \rightarrow \varphi_j(U_i \cap U_j)$$

is an element of Γ .

Definition 1.1.2.2 An **complete atlas** of M compatible with Γ is an atlas of M compatible with Γ .

Then, every atlas is contained in a unique complete atlas.

Remark Suppose A, A' be two atlases. Then, if they are **compatible**

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1.1.3 Differentiable Manifold

Definition 1.1.3.1 ...

1.1.4 Differentiable Mapping

1.1.5 Differentiable Curve, Tangent Vectors

Differentiable Curve

Definition 1.1.5.1 ...

Tangent Vectors

Definition 1.1.5.2 Let $\mathfrak{F}(p), p \in M$ be the equivalence class of differentiable functions at p ,

$$\mathfrak{F}(p) := \{(U, f) | f \in C^1(U, \mathbb{R}), p \in U \subseteq X \text{ open}\} / \sim \quad (U, f) \sim (V, g) \Leftrightarrow f|_{U \cap V} = g|_{U \cap V}$$

$\mathfrak{p}$ is an $\mathbb{R}$ -linear space.

Then, let $x : [a, b] \rightarrow M$ be a C^1 -differentiable function where

$$x(t_0) = p$$

Then, define

$$X : \mathfrak{F}(p) \rightarrow \mathbb{R}$$

$$[(U, f)] \mapsto \left(\frac{df(x(t))}{dt} \right) |_{t_0}$$

In other word, Xf is derivative of f in the direction of the curve $x(t)$ at $t = t_0$.
The function X satisfies the following properties:

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Lie Bracket

1.2 Tensor Algebra, Tensor Fields

Given in Smooth manifold.

1.3 Lie Groups

Given in Lie theory.

1.4 Fibre Bundles

1.4.1 Differentiable Principle Fibre Bundle

Definition 1.4.1.1 A differentiable principle fibre bundle over M with group G consists of $P(M, G, \pi)$, where:

- G acts freely on P on the right:

$$P \times G \rightarrow P \quad (u, a) \mapsto ua = R_a u \in P$$

- M is quotient space of P , and the canonical projection

$$\pi : P \rightarrow M$$

is differentiable;

- P is locally trivial:

We call:

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Fibre of Differentiable Principle Fibre Bundle

1.4.2 Fundamental Vector Field

Definition 1.4.2.1 Given a principle fiber bundle $P(M, G, \pi)$, the action of G on P induces a homomorphism σ of the Lie algebra $\mathfrak{g}$ of G into the Lie algebra $\mathfrak{X}(P)$ of vector fields on P .

$$\mathfrak{g} = L(G) \rightarrow \mathfrak{X}(P)$$

Constructions:

- Exponential Maps:
- Consider the action

$$\Phi_u : G \rightarrow P \quad g \mapsto u \cdot g$$

Then,

$$\sigma : \mathfrak{g} = T_e G \rightarrow \mathfrak{X}(P) \sigma(A)_u = (d\Phi_u)_e(A)$$

Take the following notation

$$A^* := \sigma A$$

Furthermore, we have

♠ **Proposition 1.4.2.2**

$$\mathfrak{g} \rightarrow T_u P \quad A \mapsto (A^*)_u$$

is linear isomorphism.

◦ **Proof**

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□

♠ **Proposition 1.4.2.3** Let A^* be fundmanetal vector field, then there exists correspondence

$$\begin{aligned} A \in \mathfrak{g} &\Rightarrow A^* \in \mathfrak{X}(P) \\ (ad(a^{-1}))A \in \mathfrak{g} &\Rightarrow (R_a)_* A^* \end{aligned}$$

◦ **Proof** ...

□

The mapping σ is injective, but not surjective. But we could define a Lie-subalgebra of $\mathfrak{X}(P)$, so that $\sigma : L(G) \rightarrow \mathfrak{X}(P)$ is bijective.

Definition 1.4.2.4 ...

♠ **Proposition 1.4.2.5** Let M be a manifold. $\{U_\alpha\}$ is an open covering of M and G a Lie group. Given mapping

◦ **Proof** ...

□

Chapter 2 Theory of Connections

2.1 Connections in a Principle Vector Bundle

Remark Firstly, the fibre through u is

$$E_{\pi(u)}$$

Definition 2.1.0.1 Let (P, M, G, π) be a principle fibre bundle over a manifold M . For each $u \in P$, $T_u(P)$ be the tangent space of P at u , and G_u be the subspace of $T_u(P)$ consisting of vectors tangent to fibre through u .

$$G_u := T_u(E_{\pi(u)}) = \text{Ker}(\pi_u)$$

- $T_u(P) = G_u \oplus Q_u$; G_u is the vertical subspace, Q_u is the horizontal subspace;

Take the following notation vX, hX .

- $Q_{ua} = (R_a)^* Q_u$; $u \in P, a \in G$
- Q depends differentiably on u ;

A **connection** is a distribution $u \mapsto Q_u$.

Remark 1 The tangent space is the formal tangent space, not rely on the open subset and the tangent line via limit.

2.1.1 Connection Form

◇ **Lemma 2.1.1.1** $\sigma_u : \mathfrak{g} \rightarrow T_u(P)$ is injective, $u \in P$, with induced linear isomorphism

$$\sigma_u : \mathfrak{g} \rightarrow V_u(P) \quad A \mapsto (A^*)_u$$

◦ **Proof** ...

□

Definition 2.1.1.2 Given a connection Γ in P . We define the 1-form P with the value in Lie algebra of G as follows: $\omega \in \Omega^1(P, \mathfrak{g})$,

$$\omega_p : T_u(P) \rightarrow \mathfrak{g} \quad X \mapsto A$$

such that $(A^*)_u = vX$.

Remark 1 $\mathfrak{g} \rightarrow G_u, A \mapsto (A^*)_u$ is a linear isomorphism.

♠ **Proposition 2.1.1.3** The connection form ω of a connection satisfies the following conditions:

- $\omega(A^*) = A$;
- $(R_a)^* \omega = \text{ad}(a^{-1})\omega$, that is

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for every $a \in G$ and every vector field X on P .

◦ **Proof** ... □

♠ **Proposition 2.1.1.4** ...

◦ **Proof** ... □

♠ **Proposition 2.1.1.5** ...

2.2 Existence and Extension of Connections

► **Theorem** ...

◦ **Proof** ... □

2.3 Parallelism

2.3.1 Horizontal Lift

Definition 2.3.1.1 Let $\tau = x(t)$ be a piecewise differentiable curve of C^1 in M . A **horizontal lift** or **lift** is a **horizontal curve**

$$\tau^* = u(t)$$

such that $x(t) = \pi(u(t))$.

Horizontal: Piecewise differentiable, the tangent vectors are horizontal.

♠ **Proposition 2.3.1.2** (Existence, Uniqueness)

◦ **Proof** ... □

2.3.2 Parallel Displacement

2.4 Holonomy Groups

Another Reference: Riemannian Holonomy Groups and Calibrated Geometry by Dominic D. Joyce

Definition 2.4.0.1 For point $x \in M$, denote

2.5 Curvature Form, and Structure Equation

Definition 2.5.0.1 Let $P(M, G, \pi)$ be a PFB and representation on G on V , with $\rho(a) \in \text{End}(V)$, $a \in G$. $\rho(ab) = \rho(a)\rho(b)$.

A **pseudotensorial form** of degree on P of type (ρ, V) is a V -valued r -form φ such that

$$R_a^* \varphi = \rho(a^{-1}) \cdot \varphi; a \in G$$

Such a form is called **tensorial form** if it is horizontal in the sense of

$$\varphi(X_1, \dots, X_i, \dots, X_r) = 0; X_i \in V_i P$$

♠ **Proposition 2.5.0.2**

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◦ **Proof** ...

□

Especially:

Definition 2.5.0.3

- The connection form ω is a pseudotensorial 1-form of type $ad(G)$;
- $D\omega$ is called **tensorial 2-form** of type $ad(G)$, denoted by Ω , is called the **curvature form** of ω .

2.5.1 Structure Equation

► **Theorem** (Structure Equation, E. Cartan) Let ω be a connection form and Ω be its curvature form, then

$$\Omega(X, Y) = d\omega(X, Y) + \frac{1}{2}[\omega(X), \omega(Y)]$$

◦ **Proof** ...

□

♠ **Proposition 2.5.1.1** ...

◦ **Proof** ...

□

2.5.2 Bianchi's Identity

► **Theorem** (Bianchi)

$$D\Omega = 0$$

◦ **Proof** ...

□

2.6 Mappings of Connections

2.7 Reduction Theorem

2.8 Holonomy Theorem

2.9 Flat Connections

2.10 Local, and Infinitesimal Holonomy Groups

2.11 Invariant Connections

Chapter 3 Linear Affine Connections

3.1 Connections in Vector Bundles

3.1.1 Fibre Metrics

3.2 Linear Connections

Chapter 4 Curvature and Space Forms

Chapter 5 Transformations

Chapter 6 Submanifolds

Chapter 7 Variation of Length Integral

Chapter 8 Homogeneous Spaces

Chapter 9 Symmetric Spaces